MAVIS

AIORI Hackathon 3 | Category: 6G & Future Networks

Experimental Dialog Correlation and Recovery Across Agent Adapters

Problem Statement: PS-021 | October 2026



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PROBLEM OVERVIEW

Abstract / Problem Description



Multi-agent AI systems communicate across frameworks (MCP, A2A), managing context independently.



Context is lost when an agent restarts or the network drops mid-task.



Retries after a disconnect can repeat operations, causing duplicate side effects.



There is currently no standardized dialog tracking to prevent this.



GOAL

Build a prototype that correlates tasks by explicit dialog IDs, tracks lifecycle states, deduplicates requests, and recovers durably after restarts.

FOUNDATION

Background



MCP

Connects an agent to tools and context.



A2A

Lets agents exchange tasks and messages with each other.



In unreliable networks, delivery is **at-least-once**: a sender retries until acknowledged, risking duplicate requests.



An **idempotent receiver** recognizes a repeated request and returns the earlier result instead of repeating the work.

Note: IETF discussions on agent protocols are emerging, so our schema is explicitly experimental.

RELATED WORK

Literature & RFC Review



IETF agentproto working group: active discussions on agent protocols (used as design inspiration).



Model Context Protocol (MCP) specification: defines how agents access tools and context.



Agent2Agent (A2A) protocol specification: defines task-based agent-to-agent messaging.



HTTP Idempotency-Key header draft: a known pattern for safe retries.*

TAKEAWAY

Existing work covers messaging or retries, but not a common dialog-state model across adapters.

** Note: The exact title for this IETF draft is "The Idempotency-Key HTTP Header Field", currently an active IETF Internet-Draft.*

ARCHITECTURE SPEC

Proposed Solution

Canonical Envelope

Standardizes messages into: dialog_id, task_id, request_id, seq, payload_hash

Mock Adapters

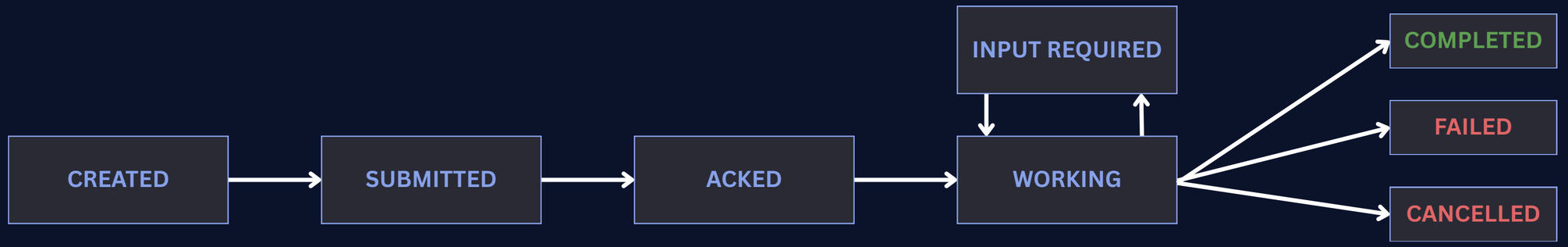
One MCP-style, one A2A-style. Both translate messages into the canonical envelope.

Core Engine

Contains a request deduplicator, lifecycle state-machine validator, and side-effect executor.

SQLite Store

A durable database holding dialogs, requests, and side effects.



Scope: Experimental prototype, not an IETF standard, no full MCP/A2A interoperability, no exactly-once claim.

RESULTS & IMPACT

Optimization Proposed

Current Status

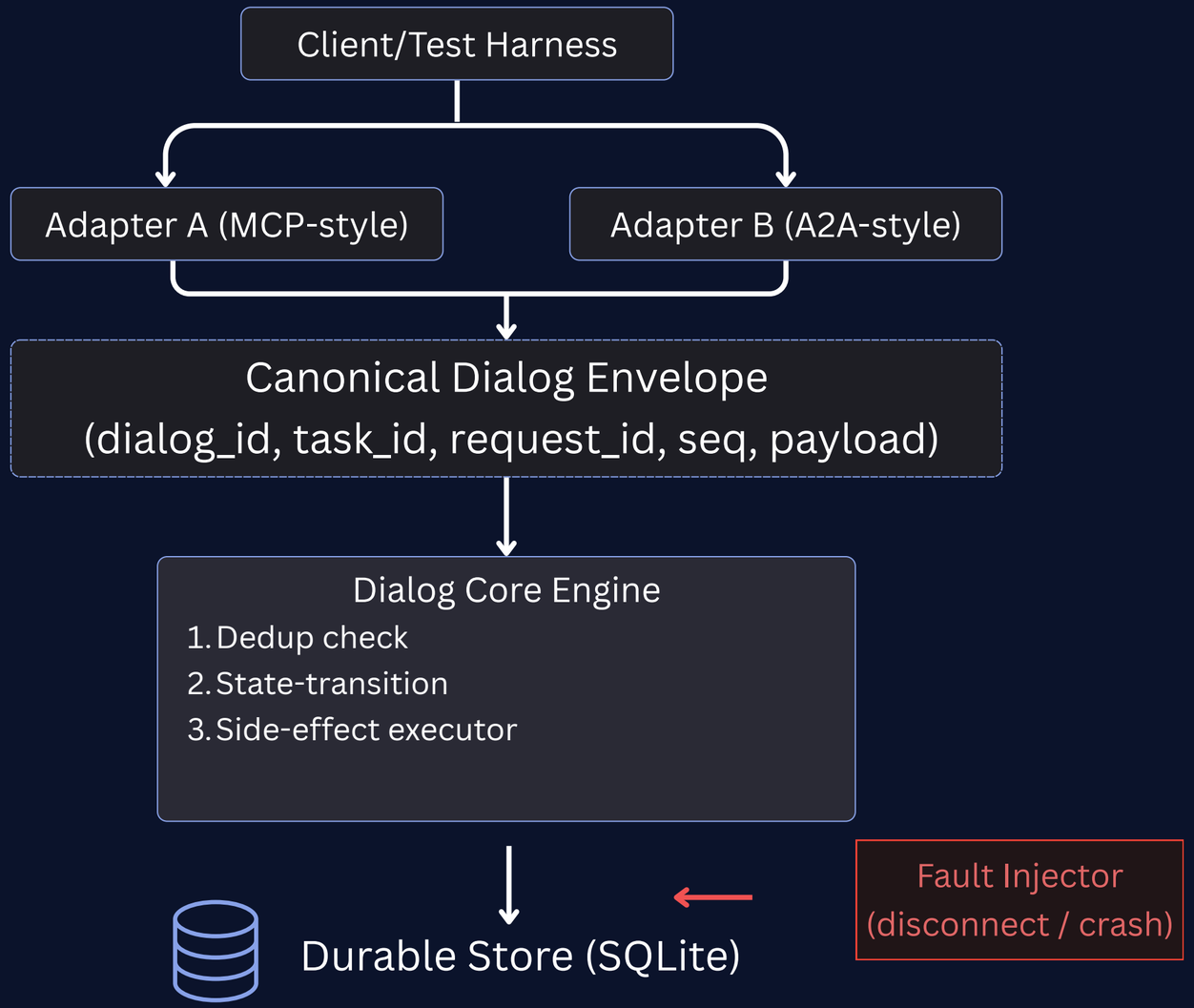
- ✗ Ephemeral context handling.
- ✗ Rampant duplicate side-effects.
- ✗ Complete operational state reset.

Optimized Architecture

- ✓ Durable state recovery mechanism.
- ✓ Robust deduplication mechanics.
- ✓ Duplicate side-effect rejection.

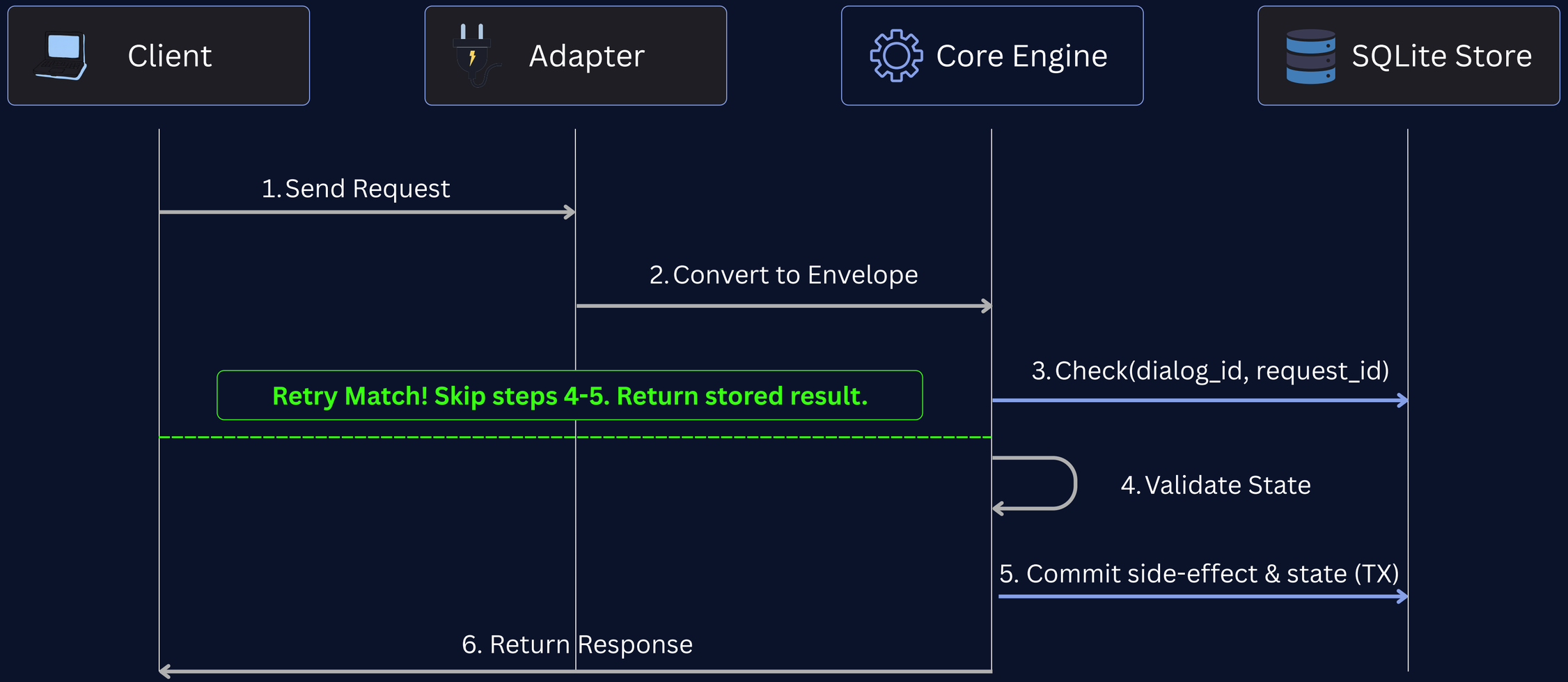
SYSTEM BLUEPRINT

Solution Architecture & Design



EXECUTION FLOW

Data Flow & Idempotency



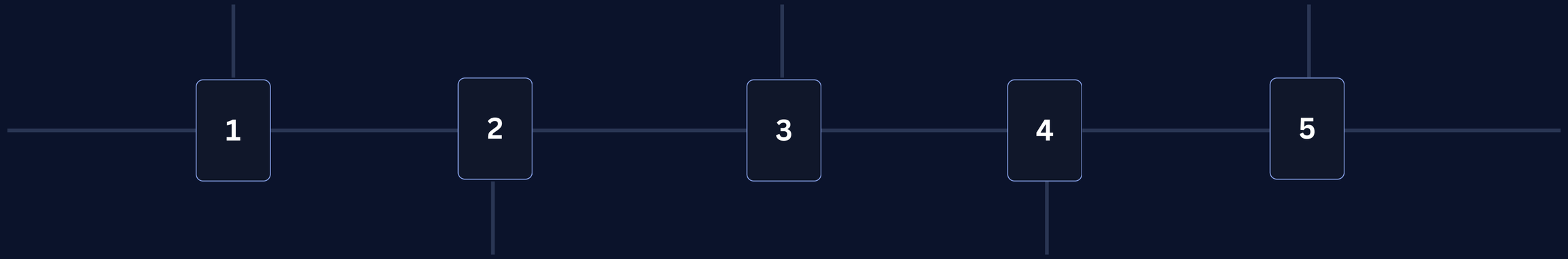
ROADMAP

Deployment & Timeline

Milestone 1
Finalize the schema and pin the version with the mentor.

Milestone 3
Both mock adapters connected to the engine.

Milestone 5
Demonstration with logs, pseudocode snippet, PDF, and repository handover.



Milestone 2
Core engine with durable store and state machine.

Milestone 4
Five disconnect/retry scenarios implemented and tested.

REFERENCE

Standards & Citations

1. IETF Agent Protocol Working Group (agentproto)
Application: Baseline for emerging IETF agent-protocol discussions and experimental schema design.

2. Anthropic Model Context Protocol (MCP)
Reference: Anthropic Open Specifications
Application: Architectural baseline for proprietary context management in Mock Adapter A.

3. Google Agent2Agent (A2A) Protocol
Reference: Google AI Frameworks
Application: Specification for inter-agent communication in Mock Adapter B.

4. AIORI-3 PS-021 Specification
Reference: Category: 6G & Future Networks Track
Application: Core engine requirements for request deduplication, state machine validation, and 5 disconnect/retry test scenarios.

Disclaimer: Experimental prototype built for AIORI-3 PS-021. No claim of final IETF standard implementation, complete MCP/A2A interoperability, or universal exactly-once delivery.

QNA SESSION

THANK YOU FOR YOUR ATTENTION!

Check out the project and repository at

GITHUB